Power — USB-C, charger, battery protection, 3V3 / 1V8 rails

File: power.kicad_sch

ESP32-P4 core — SoC, flash, clocks, core DC-DC, reset/boot

File: mcu.kicad_sch

Wi-Fi 6 / BLE — ESP32-C6-MINI-1 over SDIO (esp-hosted)

File: wifi.kicad_sch

LTE Cat-4 + GNSS — SIM7600G-H, nano-SIM, antennas

File: lte.kicad_sch

Camera (MIPI CSI-2, 2 lanes), IR illumination, IR-cut filter, PIR

File: camera_ir.kicad_sch

microSD (SDMMC slot 0, 4-bit, 3.3 V) with power switch

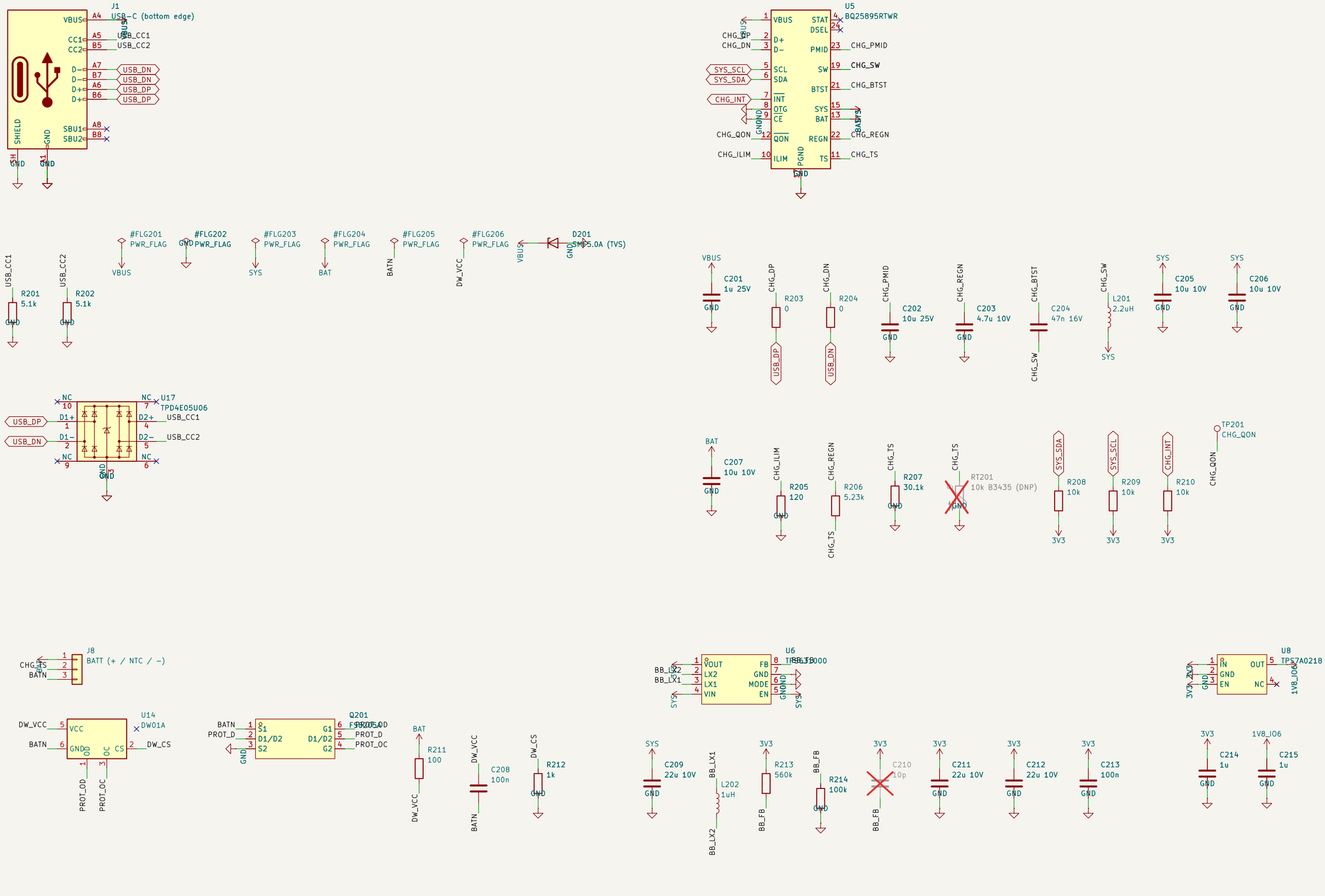
File: storage.kicad_sch

ESP32-P4 LTE Camera Board — rev A

Inter-sheet connections use global labels; power rails use power symbols (net name = value).

Generated by scripts/kisch.py from scripts/design/*.py — edit the design files, then regenerate.

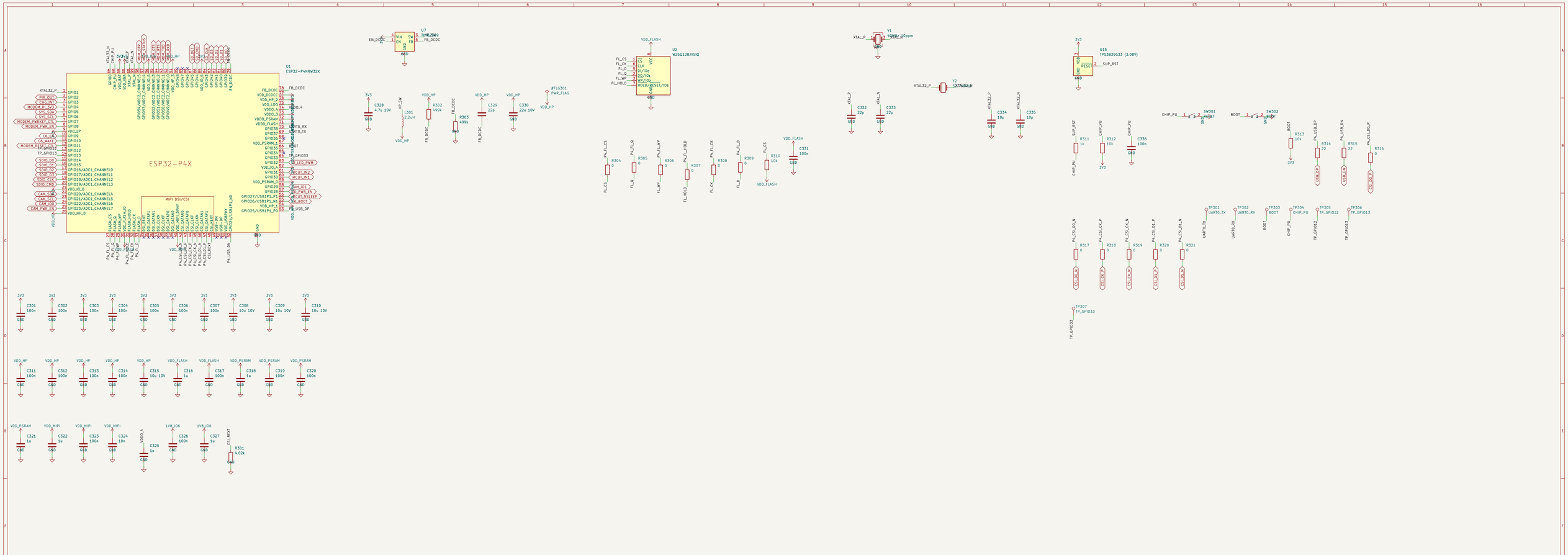
ESP32-P4 LTE Camera Board		
Portfolio design — ESP32-P4 LTE Camera		
Sheet: /		
File: esp32p4_lte_cam.kicad_sch		
Title: Top level / sheet index		
Size: A4	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/7



USB-C: sink-only, 5.1k Rd on CC1/CC2. D+/D- shared by ESP32-P4 USB-Serial/JTAG and BQ25895 BC1.2 detection.
BQ25895: ILIM 120R -> ~3 A max input; TS divider 5.23k/30.1k for 103AT NTC (JEITA) per datasheet sec. 9.2.2.
Cell protection: DW01A + FS8205A in the battery negative leg (BATN = cell −, GND = pack −).

ESP32-P4 LTE Camera Board
Portfolio design — ESP32-P4 LTE Camera
Sheet: /Power — USB-C, charger, battery protection, 3V3 / 1V8 rails/
File: power.kicad_sch
Title: Power — USB-C, charger, battery protection, 3V3 / 1V8 rails

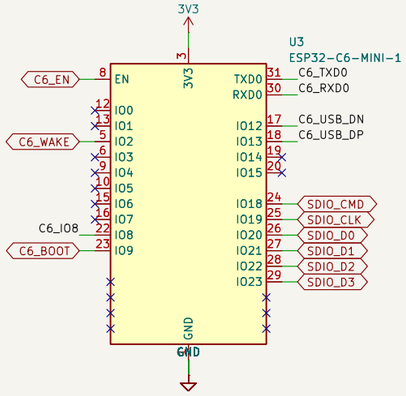
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/7

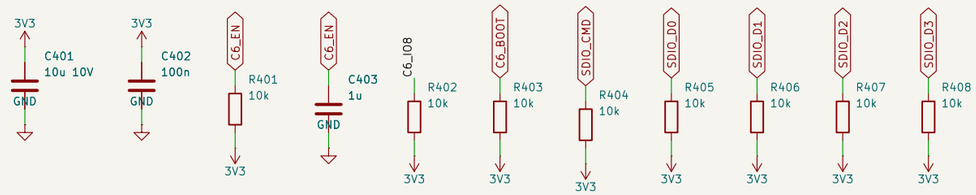


Chip rev v3.x (ESP32-P4NRW32X): VDD_HP_1 on pin 54; DC-DC feedback 499k/499k/22pF populated (HDG fig. 5).
VDD_MIPI_DPHY fed from internal VDDO_3 (2.5 V) as on ESP32-P4-Function-EV v1.5.2. VDD_USBPHY unused (HS PHY not used).
VDD_IO_6 = 1.8 V: GPIO49-54 talk to the SIM7600 UART natively. Keep these pins hi-Z while VBAT_LTE is off.

ESP32-P4 LTE Camera Board
Portfolio design — ESP32-P4 LTE Camera
Sheet: /ESP32-P4 core — SoC, flash, clocks, core DC-DC, reset/boot/
File: mcu.kicad_sch
Title: ESP32-P4 core — SoC, flash, clocks, core DC-DC, reset/boot

Size: A1	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/7

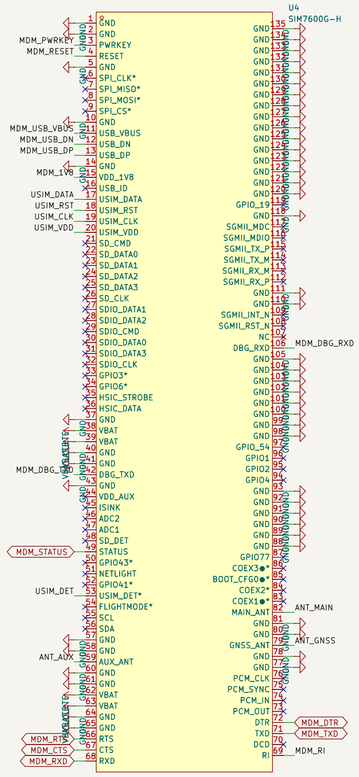


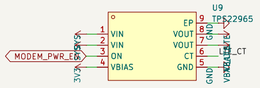


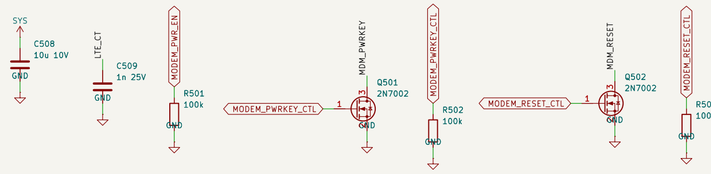


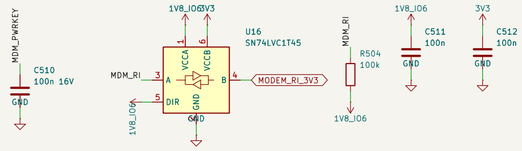
SDIO slave pins fixed on the C6 (IO18 CMD, IO19 CLK, IO20–23 D0–D3); P4 side matches ESP32-P4-Function-EV.
Antenna end of the module overhangs / keeps out at the LEFT board edge (see docs/ARCHITECTURE.md sec. 7).

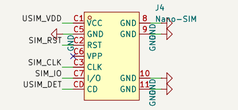
ESP32-P4 LTE Camera Board
Portfolio design – ESP32-P4 LTE Camera
Sheet: /Wi-Fi 6 / BLE – ESP32-C6-MINI-1 over SDIO (esp-hosted)/
File: wifi.kicad_sch
Title: Wi-Fi 6 / BLE – ESP32-C6-MINI-1 over SDIO (esp-hosted)
Size: A3 | Date: 2026-10-19 | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 4/7

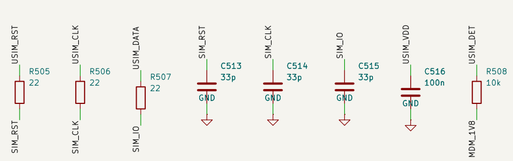


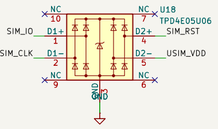


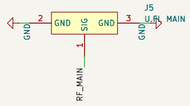


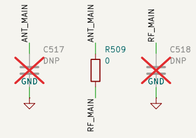


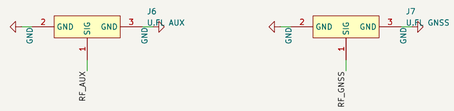


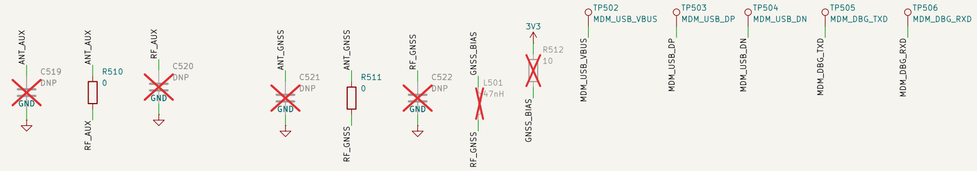






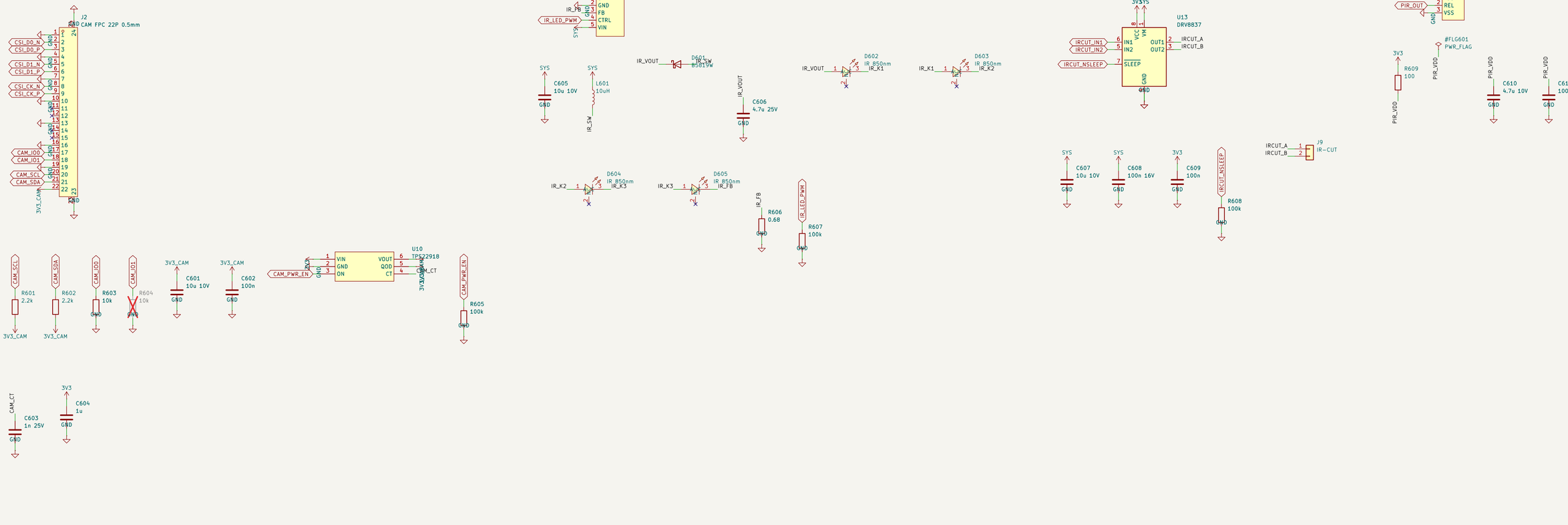






VBAT_LTE from SYS via TPS22965 (16 mOhm). >=300 uF bulk at VBAT pins per SIMCom HW design sec. 3.1 (2 A GSM bursts).
UART/STATUS/DTR are 1.8 V and connect straight to ESP32-P4 GPIO49-54 (VDD_IO_6 = 1V8_IO6).
COEX1/COEX3/BOOT_CFG0 must not be pulled up before power-up -> left NC. Pi-networks: series OR, shunts DNP.

ESP32-P4 LTE Camera Board
Portfolio design - ESP32-P4 LTE Camera
Sheet: /LTE Cat-4 + GNSS - SIM7600G-H, nano-SIM, antennas/
File: lte.kicad_sch
Title: LTE Cat-4 + GNSS - SIM7600G-H, nano-SIM, antennas
Size: A2 | Date: 2026-10-19 | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 5/7



J2: 22-pin 0.5 mm FPC, Raspberry Pi CM/Zero camera pinout; lanes 2/3 unused (ESP32-P4 CSI is 2-lane).
IR LEDs: TPS61169 constant-current boost, 4 LEDs in series, I = 0.2 V / 0.68 R = 294 mA, PWM dimming on CTRL.
Board faces the scene: PIR and IR LEDs on the top side next to the camera (front) edge.

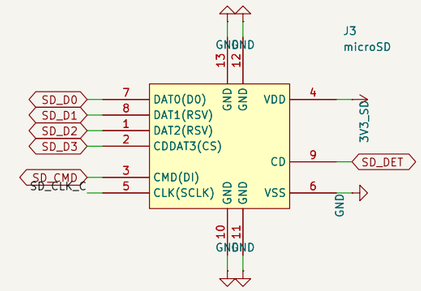
ESP32-P4 LTE Camera Board

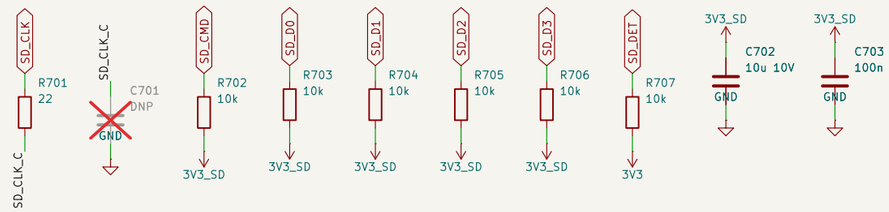
Portfolio design – ESP32-P4 LTE Camera

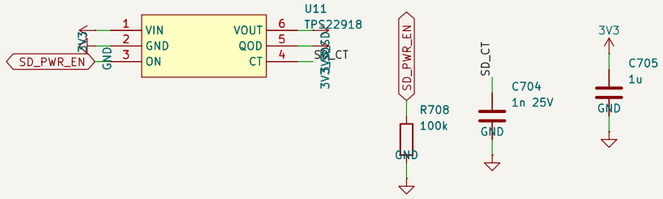
Sheet: /Camera (MIPI CSI-2, 2 lanes), IR illumination, IR-cut filter, PIR/
File: camera_ir.kicad_sch

Title: Camera (MIPI CSI-2, 2 lanes), IR illumination, IR-cut filter, PIR

Size: A2	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 6/7







SDMMC slot 0 fixed pins GPIO39–44; VDD_IO_5 at 3.3 V (no UHS-I). Pull-ups go to the switched rail 3V3_SD.

ESP32-P4 LTE Camera Board

Portfolio design – ESP32-P4 LTE Camera

Sheet: /microSD (SDMMC slot 0, 4-bit, 3.3 V) with power switch/
File: storage.kicad_sch

Title: microSD (SDMMC slot 0, 4-bit, 3.3 V) with power switch

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 7/7